

IS PRACTICAL FILE

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Suggested Practical List

1. Demonstrate the use of Network tools: ping, ipconfig, ifconfig, tracert, arp, netstat, whois.
2. Use of Password cracking tools: John the Ripper, Ophcrack. Verify the strength of passwords using these tools.
3. Use nmap/zenmap to analyze a remote machine.
4. Use Burp proxy to capture and modify the message.
5. Implement caesar cipher substitution operation.
6. Implement monoalphabetic and polyalphabetic cipher substitution operation.
7. Implement playfair cipher substitution operation.
8. Implement hill cipher substitution operation.
9. Implement rail fence cipher transposition operation.
10. Implement row transposition cipher transposition operation.
11. Implement product cipher transposition operation.

Q1. Demonstrate the use of Network tools: ping, ipconfig, ifconfig, tracert, arp, netstat, whois.

1.Ping

```
Pinging google.com [2404:6800:4002:805::200e] with 32 bytes of data:
Reply from 2404:6800:4002:805::200e: time=5ms
Reply from 2404:6800:4002:805::200e: time=7ms
Reply from 2404:6800:4002:805::200e: time=78ms
Reply from 2404:6800:4002:805::200e: time=6ms

Ping statistics for 2404:6800:4002:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 78ms, Average = 24ms
```

2.Ipconfig

```
C:\Users\ryzen>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a0d:9445:63f3:5889%16
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2405:201:4009:e861:6e53:34ea:4ce5:7cc
    Temporary IPv6 Address. . . . . : 2405:201:4009:e861:2c39:6198:1274:80aa
```

```
IPv6 Address. . . . . : 2405:201:4009:e861:6e53:34ea:4ce5:7cc
Temporary IPv6 Address. . . . . : 2405:201:4009:e861:2c39:6198:1274:80aa
Link-local IPv6 Address . . . . . : fe80::5c8d:db52:d8d1:3c66%14
IPv4 Address. . . . . : 192.168.29.114
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::4e22:f3ff:fee9:5c%14
                             192.168.29.1
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

3.Ipconfig all

```
C:\Users\ryzen>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : DESKTOP-CCLHNOT
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-10
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a0d:9445:63f3:5889%16(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 805961767
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-31-F3-5E-F8-89-D2-52-6A-2B
NetBIOS over Tcpip. . . . . : Enabled
```

Wireless LAN adapter Local Area Connection* 9:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. : FA-89-D2-52-6A-2B
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. : FE-89-D2-52-6A-2B
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter
Physical Address. : F8-89-D2-52-6A-2B
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
IPv6 Address. : 2405:201:4009:e861:6e53:34ea:4ce5:7cc(Preferred)
Temporary IPv6 Address. : 2405:201:4009:e861:2c39:6198:1274:80aa(Preferred)
Link-local IPv6 Address : fe80::5c8d:db52:d8d1:3c66%14(Preferred)
IPv4 Address. : 192.168.29.114(Preferred)
Subnet Mask : 255.255.255.0

Subnet Mask : 255.255.255.0
Lease Obtained. : Wednesday, May 13, 2026 7:33:57 PM
Lease Expires : Thursday, May 14, 2026 7:49:27 PM
Default Gateway : fe80::4e22:f3ff:fee9:5c%14
192.168.29.1
DHCP Server : 192.168.29.1
DHCPv6 IAID : 200837586
DHCPv6 Client DUID. : 00-01-00-01-30-31-F3-5E-F8-89-D2-52-6A-2B
DNS Servers : 2405:201:4009:e861::c0a8:1d01
192.168.29.1
2405:201:4009:e861::c0a8:1d01
NetBIOS over Tcpip. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : F8-89-D2-52-6A-2C
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

4.Tracert

```
C:\Users\ryzen>tracert google.com

Tracing route to google.com [2404:6800:4002:805::200e]
over a maximum of 30 hops:

  1  31 ms    2 ms     2 ms  2405:201:4009:e861:4e22:f3ff:fee9:5c
  2  *        *        *      Request timed out.
  3  36 ms    3 ms     4 ms  2405:200:801:300::71
  4  6 ms     6 ms     6 ms  2405:200:801:300::eed4
  5  85 ms    7 ms     4 ms  2405:200:805:3168:61::4
  6  *        *        *      Request timed out.
  7  *        84 ms   5 ms  2405:200:801:300::dc4
  8  *        *        *      Request timed out.
  9  80 ms   10 ms    9 ms  2405:203:400:100:172:31:5:113
 10  78 ms   5 ms     6 ms  2001:4860:1:1::2218
 11  6 ms     5 ms     6 ms  2404:6800:81e1:440::1
 12  24 ms    8 ms     6 ms  2001:4860:0:1::54ec
 13  16 ms    7 ms     7 ms  2001:4860:0:1::54f5
 14  27 ms    6 ms     6 ms  tzdeLa-be-in-x0e.1e100.net [2404:6800:4002:805::200e]

Trace complete.
```

5.Arp a

```
C:\Users\ryzen>arp -a

Interface: 192.168.29.114 --- 0xe
    Internet Address      Physical Address      Type
    192.168.29.1          4c-22-f3-e9-00-5c    dynamic
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0x10
    Internet Address      Physical Address      Type
    192.168.56.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
```

6.Netstat

```
C:\Users\ryzen>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:5354	DESKTOP-CCLHNOT:49669	ESTABLISHED
TCP	127.0.0.1:5354	DESKTOP-CCLHNOT:49670	ESTABLISHED
TCP	127.0.0.1:49669	DESKTOP-CCLHNOT:5354	ESTABLISHED
TCP	127.0.0.1:49670	DESKTOP-CCLHNOT:5354	ESTABLISHED
TCP	127.0.0.1:49672	DESKTOP-CCLHNOT:49673	ESTABLISHED
TCP	127.0.0.1:49673	DESKTOP-CCLHNOT:49672	ESTABLISHED
TCP	127.0.0.1:49674	DESKTOP-CCLHNOT:49675	ESTABLISHED
TCP	127.0.0.1:49675	DESKTOP-CCLHNOT:49674	ESTABLISHED
TCP	127.0.0.1:52165	DESKTOP-CCLHNOT:52166	ESTABLISHED
TCP	127.0.0.1:52166	DESKTOP-CCLHNOT:52165	ESTABLISHED
TCP	192.168.29.114:54136	172.188.155.25:https	ESTABLISHED
TCP	192.168.29.114:65070	10:https	ESTABLISHED
TCP	:::1:8888	DESKTOP-CCLHNOT:55488	ESTABLISHED
TCP	:::1:49762	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	:::1:53224	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	:::1:53226	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	:::1:55488	DESKTOP-CCLHNOT:8888	ESTABLISHED
TCP	:::1:61920	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	:::1:62626	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	:::1:62885	DESKTOP-CCLHNOT:8888	TIME_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:49272	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:49680	[2603:1040:a06:6::1]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:50138	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:50930	[2606:4700:90c2:7cbc:c2a6:b01:5ff2:75c4]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51109	[2600:1901:0:ac66::]:https	TIME_WAIT

TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51109	[2600:1901:0:ac66::]:https	TIME_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51438	g2600-140f-2e00-098f-0000-0000-0000-0057:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51469	[2603:1063:2001:2003::365:ff1]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51470	[2603:1063:2001:2003::365:ff1]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51473	[2603:1063:2001:1903::365:ff1]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:51495	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:53375	[2600:1901:0:ac66::]:https	TIME_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:54494	[2600:1901:0:ac66::]:https	TIME_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:55533	[2405:200:1630:68c::3544]:https	CLOSE_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:55996	[2a05:d014:1776:c001:4017:b128:30a8:cece]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:56039	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:56299	[2606:4700:90c1:d8b7:9a59:6:ccf5:ac50]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:56562	whatsapp-cdn6-shv-02-del3:5222	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:56634	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:57030	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:57271	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:58193	whatsapp-cdn6-shv-02-del3:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:59592	[2600:1901:0:ac66::]:https	TIME_WAIT
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:59805	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:59876	[2606:4700:9644:2022:f664:4:b72f:4edb]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:60158	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:60707	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:60750	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:61492	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:61521	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:61759	[2600:1901:0:ac66::]:https	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:61815	lcbomp-in-f188:5228	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:63133	sl-in-f188:5228	ESTABLISHED
TCP	[2405:201:4009:e861:2c39:6198:1274:80aa]:63860	g2600-140f-2e00-098f-0000-0000-0000-0057:https	ESTABLISHED

7. Whois

```
C:\Windows\System32\WhoIs>whois.exe google.com

Whois v1.21 - Domain information lookup
Copyright (C) 2005-2019 Mark Russinovich
Sysinternals - www.sysinternals.com

Connecting to COM.whois-servers.net...

WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-05-13T15:04:30Z <<<
```

```
Connecting to whois.markmonitor.com...

WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-08-02T02:17:33+0000
Creation Date: 1997-09-15T07:00:00+0000
Registrar Registration Expiration Date: 2028-09-13T07:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Organization: Google LLC
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Name Server: ns4.google.com
Name Server: ns3.google.com
Name Server: ns1.google.com
Name Server: ns2.google.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-05-13T15:00:29+0000 <<<
```

For more information on WHOIS status codes, please visit:

<https://www.icann.org/resources/pages/epp-status-codes>

If you wish to contact this domain's Registrant or Technical contact, and such email address is not visible above, you may do so via our web form, pursuant to ICANN's Temporary Specification. To verify that you are not a robot, please enter your email address to receive a link to a page that facilitates email communication with the relevant contact(s).

Web-based WHOIS:

<https://domains.markmonitor.com/whois/contact/google.com>

If you have a legitimate interest in viewing the non-public WHOIS details, send your request and the reasons for your request to whoisrequest@markmonitor.com and specify the domain name in the subject line. We will review that request and may ask for supporting documentation and explanation.

The data in MarkMonitor's WHOIS database is provided for information purposes, and to assist persons in obtaining information about or related to a domain name's registration record. While MarkMonitor believes the data to be accurate, the data is provided "as is" with no guarantee or warranties regarding its accuracy.

By submitting a WHOIS query, you agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to:

- (1) allow, enable, or otherwise support the transmission by email, telephone, or facsimile of mass, unsolicited, commercial advertising, or spam; or
- (2) enable high volume, automated, or electronic processes that send queries, data, or email to MarkMonitor (or its systems) or the domain name contacts (or

```
Domain Name: google.com
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-08-02T02:17:33+0000
Creation Date: 1997-09-15T07:00:00+0000
Registrar Registration Expiration Date: 2028-09-13T07:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Organization: Google LLC
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Name Server: ns4.google.com
Name Server: ns3.google.com
Name Server: ns1.google.com
Name Server: ns2.google.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-05-13T15:00:29+0000 <<<
```


URL of the ICANN WHOIS Data Problem Reporting System: <http://wdprs.internic.net/>
>>> Last update of WHOIS database: 2026-05-13T15:00:29+0000 <<<

For more information on WHOIS status codes, please visit:
<https://www.icann.org/resources/pages/epp-status-codes>

If you wish to contact this domain's Registrant or Technical contact, and such email address is not visible above, you may do so via our web form, pursuant to ICANN's Temporary Specification. To verify that you are not a robot, please enter your email address to receive a link to a page that facilitates email communication with the relevant contact(s).

Web-based WHOIS:
<https://domains.markmonitor.com/whois/contact/google.com>

If you have a legitimate interest in viewing the non-public WHOIS details, send your request and the reasons for your request to whoisrequest@markmonitor.com and specify the domain name in the subject line. We will review that request and may ask for supporting documentation and explanation.

The data in MarkMonitor's WHOIS database is provided for information purposes, and to assist persons in obtaining information about or related to a domain name's registration record. While MarkMonitor believes the data to be accurate, the data is provided "as is" with no guarantee or warranties regarding its accuracy.

By submitting a WHOIS query, you agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to:
(1) allow, enable, or otherwise support the transmission by email, telephone, or facsimile of mass unsolicited commercial advertising or spam; or

MarkMonitor reserves the right to modify these terms at any time.

By submitting this query, you agree to abide by this policy.

MarkMonitor Domain Management(TM)
Protecting companies and consumers in a digital world.

Visit MarkMonitor at <https://www.markmonitor.com>

Contact us at +1.8007459229

In Europe, at +44.02032062220

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Q2. Use of Password cracking tools: John the Ripper, Ophcrack. Verify the strength of passwords using these tools.

```
C:\Users\RYZEN>sudo apt install  
Command not found
```

```
C:\Users\RYZEN>cd Downloads
```

```
C:\Users\RYZEN\Downloads>tar -xf john-1.9.0-jumbo-1-win64.zip
```

```
C:\Users\RYZEN\Downloads>cd john-1.9.0-jumbo-1-win64\run
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run>mkdir jtr_practice
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run>cd jtr_practice
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>(echo 123456& echo  
password& echo password123& echo admin& echo letmein& echo qwerty& echo abc123& echo wel  
come& echo monkey& echo dragon) > mywordlist.txt
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>type mywordlist.tx  
t  
123456  
password  
password123  
admin  
letmein  
qwerty  
abc123  
welcome  
monkey  
dragon
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>python -c "print(''  
482c811da5d5b4bc6d497ffa98491e38')"
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>type hashes.txt  
482c811da5d5b4bc6d497ffa98491e38
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>C:\Users\RYZEN\Down  
loads\john-1.9.0-jumbo-1-win64\run\john --wordlist=mywordlist.txt --format=Raw-MD5 h  
ashes.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])  
No password hashes left to crack (see FAQ)
```

```
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>C:\Users\RYZEN\Down  
loads\john-1.9.0-jumbo-1-win64\run\john --show --format=Raw-MD5 hashes.txt  
?:password123  
1 password hash cracked, 0 left  
C:\Users\RYZEN\Downloads\john-1.9.0-jumbo-1-win64\run\jtr_practice>
```

Q3. Use nmap/zenmap to analyze a remote machine

```
C:\Users\ryzen>nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 21:03 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
135/tcp   filtered   msrpc
139/tcp   filtered   netbios-ssn
179/tcp   filtered   bgp
445/tcp   filtered   microsoft-ds
646/tcp   filtered   ldap
9929/tcp  open       nping-echo
31337/tcp open       Elite

Nmap done: 1 IP address (1 host up) scanned in 9.59 seconds
```

```
C:\Users\ryzen>nmap -p 80,443 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 21:03 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE      SERVICE
80/tcp    open       http
443/tcp   closed     https

Nmap done: 1 IP address (1 host up) scanned in 0.99 seconds
```

```
C:\Users\ryzen>nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 21:03 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.33s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open       ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered   smtp
80/tcp    open       http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered   msrpc
139/tcp   filtered   netbios-ssn
179/tcp   filtered   bgp
445/tcp   filtered   microsoft-ds
646/tcp   filtered   ldap
9929/tcp  open       nping-echo   Nping echo
31337/tcp open       tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.66 seconds
```

```

C:\Users\ryzen>nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 21:04 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered  smtp
80/tcp    open       http
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
179/tcp    filtered  bgp
445/tcp    filtered  microsoft-ds
646/tcp    filtered  ldap
9929/tcp   open       nping-echo
31337/tcp  open       Elite
Device type: general purpose|router
Running (JUST GUESSING): Linux 5.X|4.X|3.X|2.6.X (89%), MikroTik RouterOS 7.X (89%)
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:linux:linux_kernel:4.19 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:6
Aggressive OS guesses: Linux 5.0 - 5.14 (89%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (89%), Linux 4.19 (88%), Linux 4.15 - 5.19 (88%), Linux 3.10 - 4.11 (87%), Linux 4.19 - 5.15 (86%), OpenWrt 21.02 (Linux 5.4) (86%), Linux 3.2 - 4.14 (86%), Linux 2.6.32 - 3.13 (85%), Linux 5.1 - 5.15 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 25 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.89 seconds

```

```

C:\Users\ryzen>nmap -A scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 21:06 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open       ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open       http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
|_ http-server-header: Apache/2.4.7 (Ubuntu)
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
179/tcp    filtered  bgp
445/tcp    filtered  microsoft-ds
646/tcp    filtered  ldap
9929/tcp   open       nping-echo   Nping echo
31337/tcp  open       tcpwrapped
Device type: general purpose|router
Running (JUST GUESSING): Linux 5.X|4.X|3.X|2.6.X (89%), MikroTik RouterOS 7.X (89%)
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:linux:linux_kernel:4.19 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:6
Aggressive OS guesses: Linux 5.0 - 5.14 (89%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (89%), Linux 4.19 (88%), Linux

```

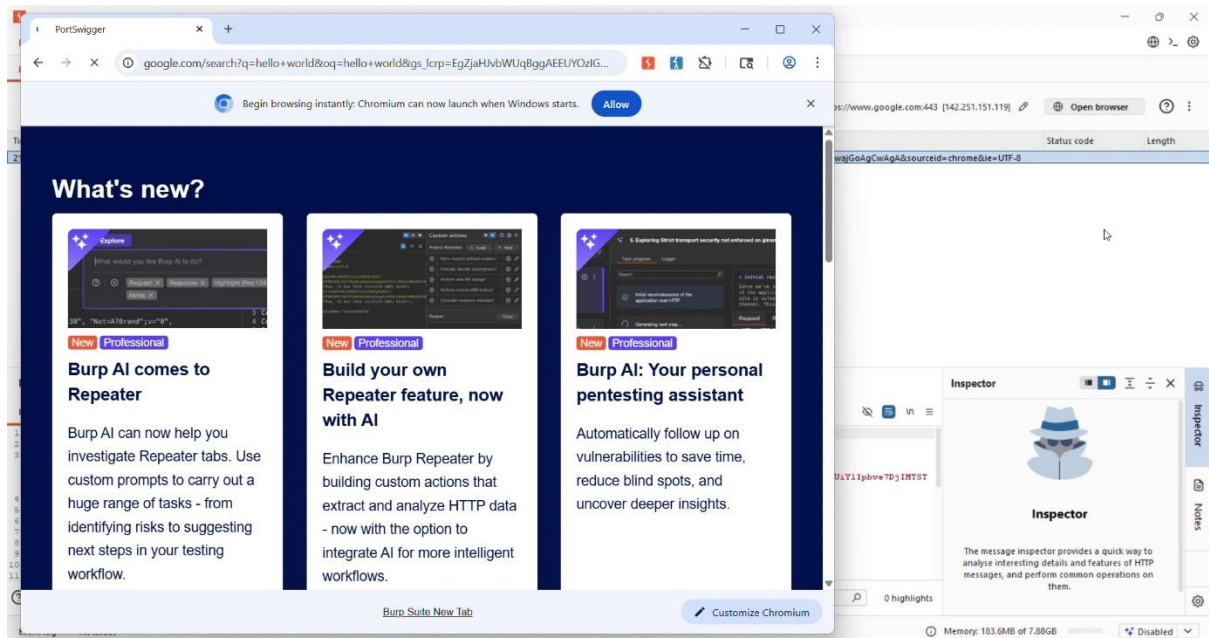
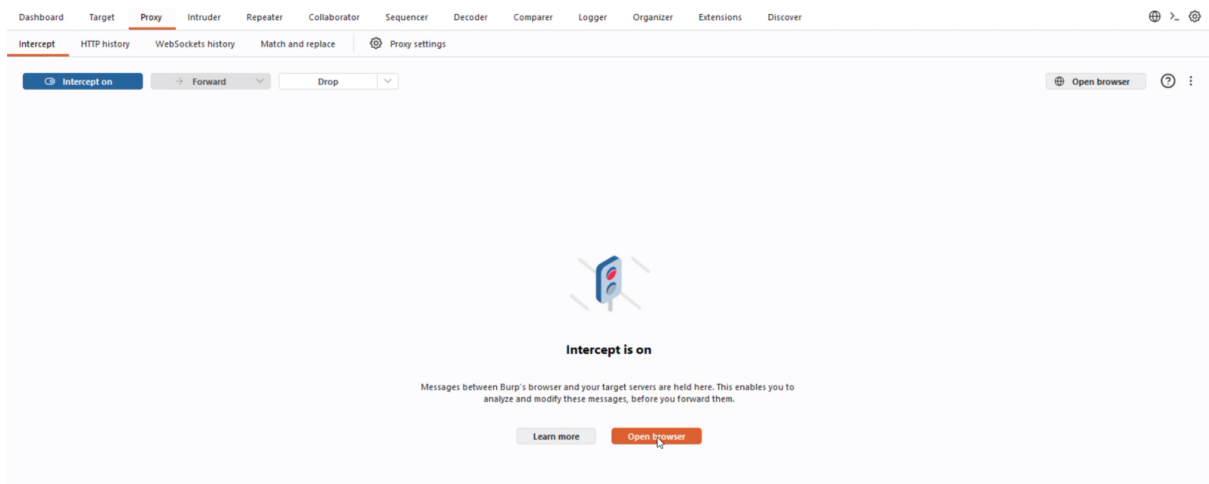
No exact OS matches for host (test conditions non-ideal).
Network Distance: 25 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

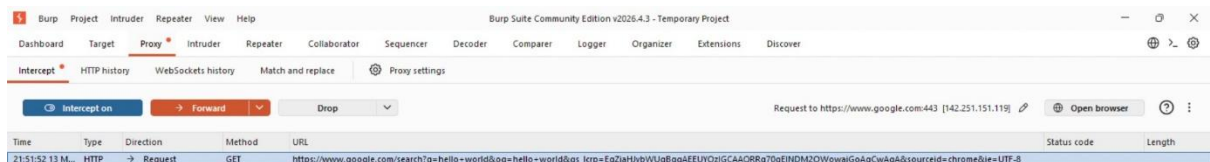
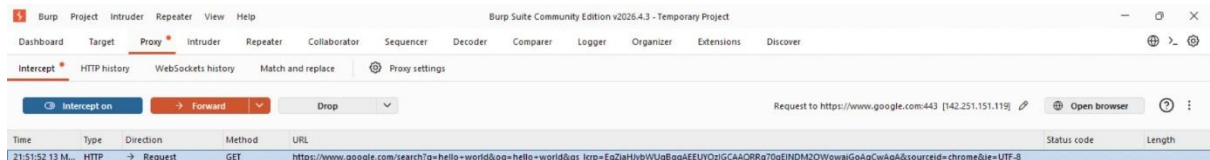
TRACEROUTE (using port 993/tcp)

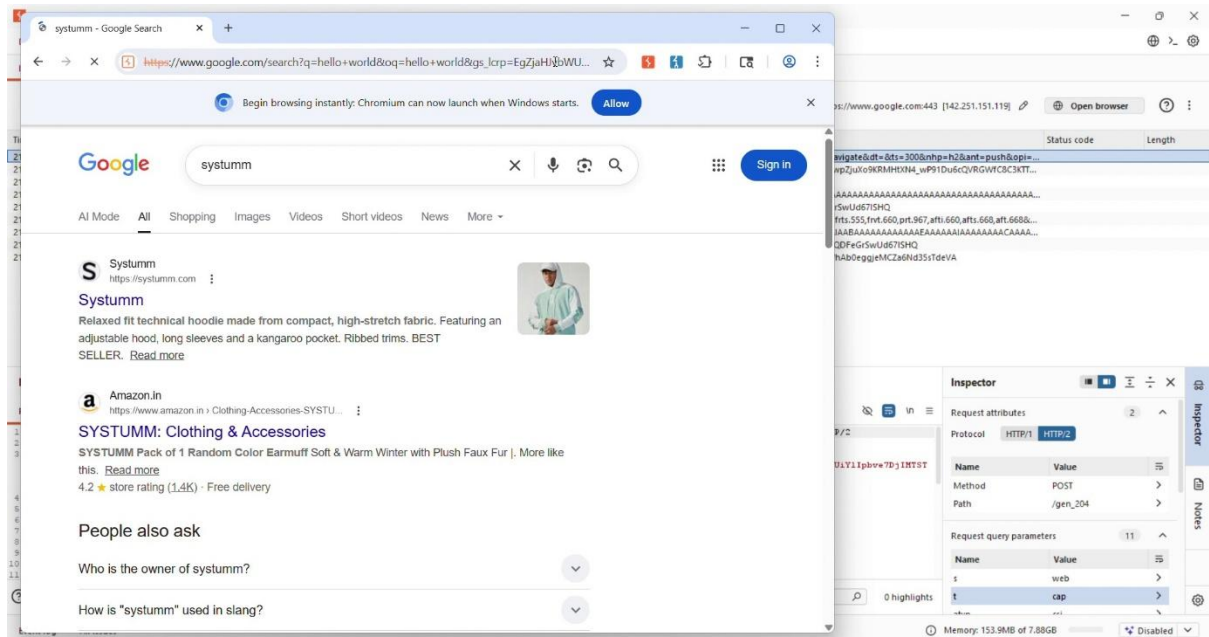
HOP	RTT	ADDRESS
1	2.00 ms	reliance.reliance (192.168.29.1)
2	7.00 ms	10.4.240.1
3	11.00 ms	172.16.26.1
4	16.00 ms	192.168.128.138
5	20.00 ms	172.27.248.53
6	15.00 ms	172.27.248.35
7	43.00 ms	192.168.44.22
8	...	10
11	39.00 ms	103.198.140.176
12	255.00 ms	103.198.140.205
13	249.00 ms	ae-21.a02.nycmny17.us.bb.gin.ntt.net (128.241.7.158)
14	347.00 ms	ae-5.r23.nwrknj03.us.bb.gin.ntt.net (129.250.3.50)
15	219.00 ms	ae-14.a00.nwrknj03.us.bb.gin.ntt.net (129.250.3.177)
16	293.00 ms	ae-1.akamai.nwrknj03.us.bb.gin.ntt.net (128.241.10.113)
17	274.00 ms	ae1.r01.ewr01.icn.netarch.akamai.com (23.203.154.34)
18	278.00 ms	ae19.r01.ord01.icn.netarch.akamai.com (23.193.113.37)
19	274.00 ms	ae16.r01.sjc01.icn.netarch.akamai.com (23.32.62.79)
20	292.00 ms	ae1.r12.sjc01.iem.netarch.akamai.com (23.207.232.37)
21	282.00 ms	ae22.gw4.scz1.netarch.akamai.com (23.203.158.53)
22	...	24
25	272.00 ms	scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 40.11 seconds

Q4 Use Burp proxy to capture and modify the message.







Q5 Implement caesar cipher substitution operation.

```
In [1]: def caesar_encrypt(text, shift):
        result = ""
        for char in text:
            if 'a' <= char <= 'z':
                result += chr(((ord(char) - ord('a') + shift) % 26) + ord('a'))
            elif 'A' <= char <= 'Z':
                result += chr(((ord(char) - ord('A') + shift) % 26) + ord('A'))
            else:
                result += char
        return result
```

```
In [2]: def caesar_decrypt(text, shift):
        return caesar_encrypt(text, -shift)
```

```
In [3]: text = input("Text: ")
        shift = int(input("Shift: "))

        enc = caesar_encrypt(text, shift)
        dec = caesar_decrypt(enc, shift)

        print("Encrypted:", enc)
        print("Decrypted:", dec)
```

Encrypted: KHOOR ZRUOG

Decrypted: HELLO WORLD

```
In [ ]:
```

Q6 Implement monoalphabetic and polyalphabetic cipher substitution operation.

Monoalphabetic Cipher

```
In [1]: def monoalphabetic_encrypt(plaintext, key):
    encrypted_text = ""

    alphabet = 'abcdefghijklmnopqrstuvwxyz'
    mapping = str.maketrans(alphabet, key)

    for char in plaintext:
        if char.isalpha():
            if char.islower():
                encrypted_text += char.translate(mapping)
            else:
                encrypted_text += char.lower().translate(mapping).upper()
        else:
            encrypted_text += char
    return encrypted_text

def monoalphabetic_decrypt(ciphertext, key):
    decrypted_text = ""

    alphabet = 'abcdefghijklmnopqrstuvwxyz'
    reverse_mapping = str.maketrans(key, alphabet)

    for char in ciphertext:
        if char.isalpha():
            if char.islower():
                decrypted_text += char.translate(reverse_mapping)
            else:
                decrypted_text += char.lower().translate(reverse_mapping).upper()
        else:
            decrypted_text += char
    return decrypted_text

# Example Usage - Monoalphabetic Cipher
mono_plaintext = "Hello World"
mono_key = "zyxwvutsrqponmlkjihgfedcba" # A simple reverse alphabet key

mono_encrypted = monoalphabetic_encrypt(mono_plaintext, mono_key)
mono_decrypted = monoalphabetic_decrypt(mono_encrypted, mono_key)

print(f"Monoalphabetic Plaintext: {mono_plaintext}")
print(f"Monoalphabetic Key: {mono_key}")
print(f"Monoalphabetic Encrypted: {mono_encrypted}")
print(f"Monoalphabetic Decrypted: {mono_decrypted}")
```

```
Monoalphabetic Plaintext: Hello World
Monoalphabetic Key: zyxwvutsrqponmlkjihgfedcba
Monoalphabetic Encrypted: Suool Dliow
Monoalphabetic Decrypted: Hello World
```

Polyalphabetic Cipher (Vigenere Cipher)

```
In [2]: def polyalphabetic_encrypt(plaintext, key):
    encrypted_text = []
    key_len = len(key)
    key_as_int = [ord(char) for char in key.lower()]

    for i, char in enumerate(plaintext):
        if char.isalpha():
            shift = key_as_int[i % key_len] - ord('a')
            start = ord('a') if char.islower() else ord('A')
            encrypted_char_code = start + (ord(char) - start + shift) % 26
            encrypted_text.append(chr(encrypted_char_code))
        else:
            encrypted_text.append(char)
    return "".join(encrypted_text)

def polyalphabetic_decrypt(ciphertext, key):
    decrypted_text = []
    key_len = len(key)
```

```

key_as_int = [ord(char) for char in key.lower()]

for i, char in enumerate(ciphertext):
    if char.isalpha():
        shift = key_as_int[i % key_len] - ord('a')
        start = ord('a') if char.islower() else ord('A')
        decrypted_char_code = start + (ord(char) - start - shift + 26) % 26 # +26 to handle negative results of modulo
        decrypted_text.append(chr(decrypted_char_code))
    else:
        decrypted_text.append(char)
return "".join(decrypted_text)

poly_plaintext = "Attack at dawn"
poly_key = "Lemon"

poly_encrypted = polyalphabetic_encrypt(poly_plaintext, poly_key)
poly_decrypted = polyalphabetic_decrypt(poly_encrypted, poly_key)

print(f"\nPolyalphabetic Plaintext: {poly_plaintext}")
print(f"Polyalphabetic Key: {poly_key}")
print(f"Polyalphabetic Encrypted: {poly_encrypted}")
print(f"Polyalphabetic Decrypted: {poly_decrypted}")

```

```

Polyalphabetic Plaintext: Attack at dawn
Polyalphabetic Key: Lemon
Polyalphabetic Encrypted: Lxfopv mh oeib
Polyalphabetic Decrypted: Attack at dawn

```

In []:

Q7 Implement playfair cipher substitution operation.

```
In [1]: def create_matrix(key):
    key = key.lower().replace('j', 'i')

    filtered_key = ""
    for char in key:
        if 'a' <= char <= 'z':
            filtered_key += char

    alphabet = "abcdefghijklmnopqrstuvwxyz"

    key_string = ""

    for ch in filtered_key:
        if ch not in key_string:
            key_string += ch

    for ch in alphabet:
        if ch not in key_string:
            key_string += ch
        if len(key_string) == 25:
            break

    matrix = []
    for i in range(0, 25, 5):
        matrix.append(list(key_string[i:i+5]))

    return matrix

def find_position(matrix, ch):
    if ch == 'j':
        ch = 'i'
    for i in range(5):
        for j in range(5):
            if matrix[i][j] == ch:
                return i, j

def playfair_encrypt(text, key):
    matrix = create_matrix(key)
    text = text.replace('j', 'i')

    pairs = []
    i = 0
    while i < len(text):
        a = text[i]
        b = text[i+1] if i+1 < len(text) else 'x'

        if a == b:
            pairs.append((a, 'x'))
            i += 1
        else:
            pairs.append((a, b))
            i += 2

    result = ""

    for a, b in pairs:
        r1, c1 = find_position(matrix, a)
        r2, c2 = find_position(matrix, b)

        if r1 == r2:
            result += matrix[r1][(c1+1) % 5]
            result += matrix[r2][(c2+1) % 5]

        elif c1 == c2:
            result += matrix[(r1+1) % 5][c1]
            result += matrix[(r2+1) % 5][c2]

        else:
            result += matrix[r1][c2]
            result += matrix[r2][c1]

    return result
```

```

def playfair_decrypt(ciphertext, key):
    matrix = create_matrix(key)

    if len(ciphertext) % 2 != 0:
        print("Warning: Ciphertext has odd length, which is unusual for Playfair. Padding with 'x'.")
        ciphertext += 'x'

    pairs = []
    for i in range(0, len(ciphertext), 2):
        pairs.append((ciphertext[i], ciphertext[i+1]))

    result = ""

    for a, b in pairs:
        r1, c1 = find_position(matrix, a)
        r2, c2 = find_position(matrix, b)

        if r1 == r2:
            result += matrix[r1][(c1 - 1) % 5]
            result += matrix[r2][(c2 - 1) % 5]

        elif c1 == c2:
            result += matrix[(r1 - 1) % 5][c1]
            result += matrix[(r2 - 1) % 5][c2]

        else:
            result += matrix[r1][c2]
            result += matrix[r2][c1]

    return result

key = input("Enter key: ").lower().replace(" ", "")
plaintext = input("Enter plaintext: ").lower().replace(" ", "")

print(f"Original plaintext: {plaintext}")

encrypted_text = playfair_encrypt(plaintext, key)
print(f"Encrypted text: {encrypted_text}")

decrypted_text = playfair_decrypt(encrypted_text, key)
print(f"Decrypted text: {decrypted_text}")

```

Original plaintext: helloworld
 Encrypted text: kcnvmpymqmcy
 Decrypted text: helxloworldx

Q8 Implement hill cipher substitution operation.

```
In [1]: import numpy as np

def modInverse(a, m):
    for x in range(1, m):
        if ((a % m) * (x % m)) % m == 1:
            return x
    return -1

def hill_encrypt(text, key):
    text = text.lower().replace(" ", "")
    if len(text) % 2 != 0:
        text += 'x'
    result = ""
    for i in range(0, len(text), 2):
        x = ord(text[i]) - ord('a')
        y = ord(text[i+1]) - ord('a')
        c1 = (key[0][0]*x + key[0][1]*y) % 26
        c2 = (key[1][0]*x + key[1][1]*y) % 26
        result += chr(c1 + ord('a'))
        result += chr(c2 + ord('a'))
    return result

def hill_decrypt(cipher, key):

    det = (key[0][0]*key[1][1] - key[0][1]*key[1][0]) % 26
    inv_det = modInverse(det, 26)
    if inv_det == -1:
        return "Error: Matrix is not invertible modulo 26"

    inv_key = [
        [(key[1][1] * inv_det) % 26, (-key[0][1] * inv_det) % 26],
        [(-key[1][0] * inv_det) % 26, (key[0][0] * inv_det) % 26]
    ]

    result = ""
    for i in range(0, len(cipher), 2):
        x = ord(cipher[i]) - ord('a')
        y = ord(cipher[i+1]) - ord('a')
        p1 = (inv_key[0][0]*x + inv_key[0][1]*y) % 26
        p2 = (inv_key[1][0]*x + inv_key[1][1]*y) % 26
        result += chr(int(p1) + ord('a'))
        result += chr(int(p2) + ord('a'))
    return result

choice = input("Use hardcoded key (3, 3, 2, 5)? (y/n): ").lower()
if choice == 'y':
    key = [[3, 3], [2, 5]]
else:
    print("Enter 4 values for 2x2 matrix (e.g., 3 3 2 5):")
    a, b, c, d = map(int, input().split())
    key = [[a, b], [c, d]]

text = input("Enter text: ")
encrypted = hill_encrypt(text, key)
print("Encrypted:", encrypted)
decrypted = hill_decrypt(encrypted, key)
print("Decrypted:", decrypted)
```

Encrypted: hiozeipjql

Decrypted: helloworld

Q9 Implement rail fence cipher transposition operation.

```
In [1]: def rail_fence_encrypt(text, key):
    rail = [['\n' for _ in range(len(text))] for _ in range(key)]

    row, col = 0, 0
    down = True

    for ch in text:
        rail[row][col] = ch
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    result = []
    for i in range(key):
        for j in range(len(text)):
            if rail[i][j] != '\n':
                result.append(rail[i][j])
    return ''.join(result)

def rail_fence_decrypt(ciphertext, key):

    rail = [['\n' for _ in range(len(ciphertext))] for _ in range(key)]

    row, col = 0, 0
    down = True
    for i in range(len(ciphertext)):
        rail[row][col] = '*'
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    index = 0
    for i in range(key):
        for j in range(len(ciphertext)):
            if rail[i][j] == '*':
                rail[i][j] = ciphertext[index]
                index += 1

    result = []
    row, col = 0, 0
    down = True
    for i in range(len(ciphertext)):
        result.append(rail[row][col])
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    return ''.join(result)
```

```
text = input("Enter plaintext: ")
key = int(input("Enter number of rails: "))

encrypted_text = rail_fence_encrypt(text, key)
print("Encrypted:", encrypted_text)

decrypted_text = rail_fence_decrypt(encrypted_text, key)
print("Decrypted:", decrypted_text)
```

Encrypted: HLOWRDEL OL

Decrypted: HELLO WORLD

Q10 Implement row transposition cipher transposition operation.

```
In [1]: def encryption_function(text, key):
text = text.replace(" ", "")
cols = len(key)
rows = (len(text) + cols - 1) // cols
matrix = [['X'] * cols for _ in range(rows)]
k = 0
for i in range(rows):
    for j in range(cols):
        if k < len(text):
            matrix[i][j] = text[k]
            k += 1
result = ""
for num in range(1, cols+1):
    col_index = key.index(num)
    for row in matrix:
        result += row[col_index]
return result
```

```
In [2]: def decryption_function(cipher, key):
cols = len(key)
rows = len(cipher) // cols
matrix = [[''] * cols for _ in range(rows)]
curr = 0
for num in range(1, cols + 1):
    col_index = key.index(num)
    for r in range(rows):
        matrix[r][col_index] = cipher[curr]
        curr += 1
plain = ""
for r in range(rows):
    for c in range(cols):
        plain += matrix[r][c]
return plain
```

```
In [3]: text = input("text: ")
key = list(map(int, input("key: ").split()))

enc = encryption_function(text, key)
print("Encoded:", enc)

dec = decryption_function(enc, key)
print("Decoded:", dec)
```

Encoded: ewdlrXholloX
Decoded: helloworldXX

In []:

Q11 Implement product cipher transposition operation.

```
In [1]: def caesar(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            result += chr((ord(ch.upper()) - ord('A') + shift) % 26 + ord('A'))
    return result

def caesar_decrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            result += chr((ord(ch.upper()) - ord('A') - shift) % 26 + ord('A'))
    return result

def rail_fence(text, key):
    rail = ['' for _ in range(key)]
    row, down = 0, True
    for ch in text:
        rail[row] += ch
        if row == 0: down = True
        elif row == key - 1: down = False
        row += 1 if down else -1
    return ''.join(rail)

def rail_fence_decrypt(cipher, key):
    idx_rail = [[] for _ in range(key)]
    row, down = 0, True
    for i in range(len(cipher)):
        idx_rail[row].append(i)
        if row == 0: down = True
        elif row == key - 1: down = False
        row += 1 if down else -1

    result = [''] * len(cipher)
    current_char_idx = 0
    for r in range(key):
        for pos in idx_rail[r]:
            result[pos] = cipher[current_char_idx]
            current_char_idx += 1
    return ''.join(result)

def encryption(text, shift, rails):
    step1 = caesar(text, shift)
    step2 = rail_fence(step1, rails)
    return step2

def decryption(cipher, shift, rails):
    step1 = rail_fence_decrypt(cipher, rails)
    step2 = caesar_decrypt(step1, shift)
    return step2
```

```
In [2]: text = input("Enter plaintext: ")
    shift = int(input("Enter shift: "))
    rails = int(input("Enter rails: "))

    encrypted = encryption(text, shift, rails)
    print("Encrypted:", encrypted)

    decrypted = decryption(encrypted, shift, rails)
    print("Decrypted:", decrypted)
```

Encrypted: JQFGNYTNQ
Decrypted: HELLOWORD